

Vishwashankar T. Janakiraman

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Summary

Data Scientist/Engineer with 5+ years in implementing machine learning, large language models, developing scalable ETL workflows and optimizing data storage solutions. Expertise in PySpark, ByteCloud, AWS, Databricks, Snowflake and Graph Analytics. Proven ability to deliver data-driven solutions, enhance performance, and streamline operations.

Experience

TikTok

New York City, NY, USA

Senior Data Scientist/Developer (ByteCloud, PySpark, Docker, Dataleap)

Feb 2024 – Present

Financial Crime Compliance

- Designed and deployed a **community detection** algorithm to cluster over **150M+ users**, IP addresses, and government-issued IDs, improving financial crime investigation coverage by 65% and reducing investigation time by 30%.
- Built a custom graph traversal algorithm to surface high-risk sub-networks among suspicious entities, accelerating triage for flagged accounts and contributing to a 45% increase in detection of coordinated fraud patterns.
- Developed and productionized **anomaly detection** models using **Isolation Forest**, assigning dynamic risk scores to **500M+** users, transactions/month in the TikTok Live network, increasing outlier detection precision by 35% and reducing false positives by 20%.
- Deployed LLM models and vector embeddings using ByteCloud containers on **NVIDIA A100 GPUs**, exposing them as **low-latency API** endpoints used by internal tools, supporting 10K queries/day.
- Built a Retrieval-Augmented Generation (**RAG**) **LLM chatbot** to generate automated investigative reports from scored and traversed high-risk networks, enabling **real-time** querying and cutting manual report generation time by 90%, saving analysts and investigators ~ 20 hours/week.
- Automated sensitive data scanning across 2M+ internal databases, reducing manual compliance reporting efforts by 80 hours/month and ensuring continuous monitoring for regulatory risks.

ADP Inc.

Parsippany, NJ, USA

Data Scientist/Engineer (AWS, Databricks, Snowflake, Airflow)

July 2021 – Feb 2024

- Designed and developed complex **ETL** workflows, **data pipelines** using Databricks, **AWS** State Machines, and Glue to streamline data processing from diverse sources into a centralized data lake and performed thorough end-to-end testing.
- Automated ETL orchestration with AWS Step Functions, S3 Aurora tables, DynamoDB. Optimizing performance, efficiency by **70%**
- Built real-time **text classification** models using AWS **SageMaker** for earnings, deductions, and other use cases, integrating automated **ML** classification endpoints.
- Enhanced storage efficiency with data partitioning strategies in S3 and Aurora, and implemented automated data quality checks in PySpark-**Databricks** to proactively detect anomalies.
- Collaborated with cross-functional teams to translate business requirements into technical solutions, reducing costs and improving data testing workflows.

Data Science Intern (PostgreSQL, NetworkX, Plotly Dashboard, AWS)

June 2020 – Aug 2020

- Optimized the caching strategy of a data retrieval system by using **NLP** techniques on unstructured network queries.
- Performed advanced analytics using Python to obtain network and user features.
- Designed prototypes and obtained additional features using Graph Clustering and **Community Detection** methods for inferring network and user statistics.
- Performed extensive data analysis to develop **Graph Clustering** and Community Detection algorithms for inferring network and user statistics.

Infosys Ltd.

Mysore, KA, India

Software Developer Intern (Angular, Spring, OracleDB)

Dec 2018 – May 2019

- Developed a web application for the security department to handle employee complaints, integrating OracleDB, Spring, **REST APIs**, and **Angular 4**.
- Delivered the project within **80%** of the proposed duration using agile methodology, significantly enhancing complaint resolution efficiency.

Technical Skills

Languages: Python, PySpark, R, Bash/sh, Java, SQL

Data Services: Databricks, AWS, Dataleap, Snowflake, Plotly, Tableau.

Database: MongoDB, DynamoDB, PostgreSQL, MySQL, OracleDB, Abase, Clickhouse, Redis, Neo4j, ElasticSearch.

Graph Analytics: Community Detection, Node Embedding, Graph Convolutional Neural Networks(GCNs), Variational AutoEncoders.

ML/DL: Supervised and Unsupervised Models, PCA, Feature Engineering, CNNs, Logistic Regression.

Education

New Jersey Institute of Technology

May 2021

Master of Science in Data Science

GPA: 3.75/4.0

Amrita Vishwa Vidyapeetham University

May 2019

Bachelors of Technology in Computer Science and Engineering (CSE)

GPA: 3.8/4.0